GTU CSE 241 / 501 Quiz

Std Code:1806

```
#ifndef container_h
#define container_h
using namespace std;
#include <iostream>
#include <vector>
#include <cstdlib>
class Container{

  public:
  List<int>data;
  int getSize(){return data.size();}
  virtual bool empty()=0;
  virtual void add(int value)=0;
  virtual int remove()=0;
  friend class List;
};

class List{
protected:
 vector<int>data;
public:
 void push_back(int x){
 data.push_back(int x);
}
 void pop-back() { if(data.size()==0)exit(-1);
  data.pop-back();//delete last element
 }
 void push-front(int x){
  for(int i=0; i<data.size(); i++){
   data[i+1]=data[i];
  }
  data[0]=x;
 }
 void pop_front(){ if(data.size()==0 exit(-1);
  for(int i=0; i<data.size(); i++){
  data[i]=data[i+1];}
  data[data.size()-1]=0;}
 }; //close
class Stack:public Container {
public:
 bool empty()override{

  if(data.size()==0) return true;
  else{return false;}
}
```

```
void add(int value) override {

  push_back(value);
}

int remove()override{

  pop-back(value);
}
}; //close

class Queue :public Container{
public:
  void add (int value)override{
   push-front(value);
  }
  int remove() override{
   pop-front();
  }
}; //close

#endif

  void delete(vector<Container*>){if(data.size()==0)exit(-1);
   for(int i=0; i< get size(); i++){
    data[i]=0;
   }
  }
int main(){

Container* c1, c2;

Stack x;
Queue y;
c1=&x;
c2=&y;

x.add(8);
x.add(9);
y.add(1);
x.remove();
y.remove();

 deleteC (data);

 return 0;

 }
```